

Lab Report Week 8

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Question 3

Consider two RVs, $X \sim \text{Bin}(n, p)$, with $m = 10$, $n = 20$, $p = 1/3$. We investigate the distribution of the sum $W = X + Y$, in particular how close the empirical probability distribution (obtained via simulation) comes to the theoretical probability distribution $\text{Bin}(n + m, p)$.

```
set.seed(111)
```

```
m <- 10
n <- 20
p <- 1/3
```

Part a)

Set $B=100$. Generate a random sample of size B from $\text{Bin}(m, p)$ and store the result in X . Similarly, generate a random sample of size B from $\text{Bin}(n, p)$ and store the result in Y . Compute W and tabulate the values using `table(W)`.

```
set.seed(111)
```

```
# 100 simulations
B <- 100
X <- rbinom(B, m, p)
Y <- rbinom(B, n, p)
W <- X + Y
table(W)
```

```
## W
##  4  5  6  7  8  9 10 11 12 13 14 15
##  4  1  3  9 11 13 15  9 11 15  5  4
```

Part b)

Convert the column names of the table to numeric, store in `W.obs`, and compute the empirical frequencies of different values of W , store in `P.obs`:

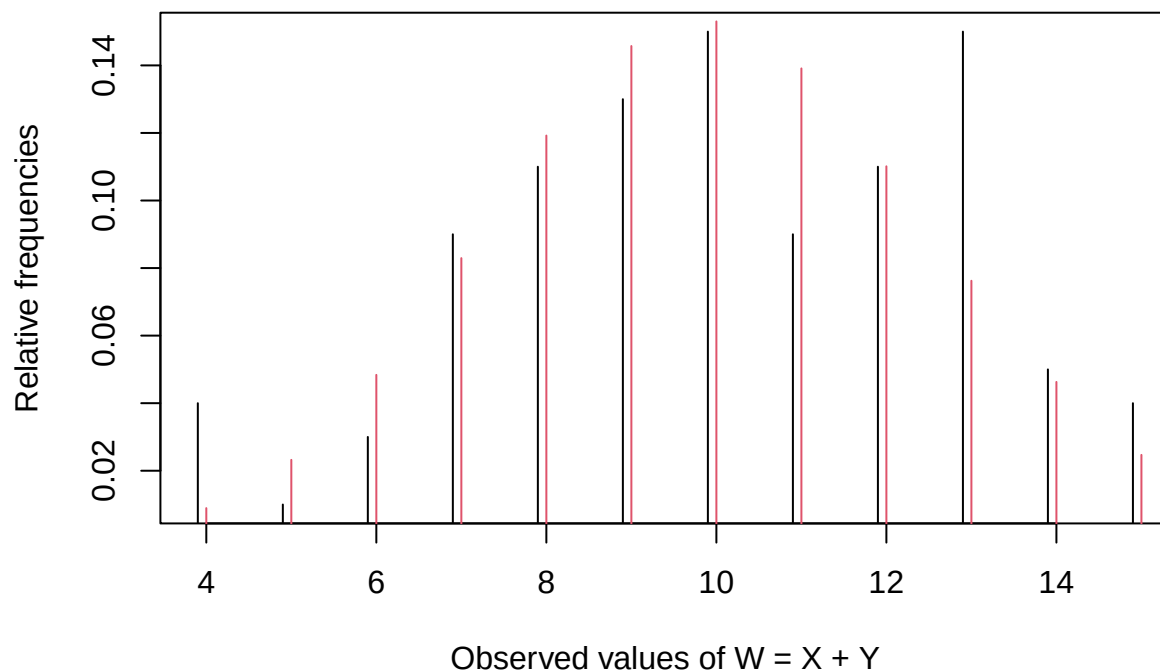
```
Tab = table(W)
W.obs = as.numeric(names(Tab))
P.obs = as.numeric(Tab/B)
```

```
Tab = table(W)
W.obs = as.numeric(names(Tab))
P.obs = as.numeric(Tab/B)
```

Part c)

Plot the empirical frequencies and the theoretical probabilities from $Bin(n + m, p)$ (obtained using `dbinom(0:(m+n), m+n, p=1/3)`), and comment on how close they are.

```
plot(W.obs - 0.1, P.obs, type='h',  
     xlab = "Observed values of W = X + Y",  
     ylab = "Relative frequencies")  
points(0:(m+n), dbinom(0:(m+n), m + n, p=1/3), type='h', col=2)
```



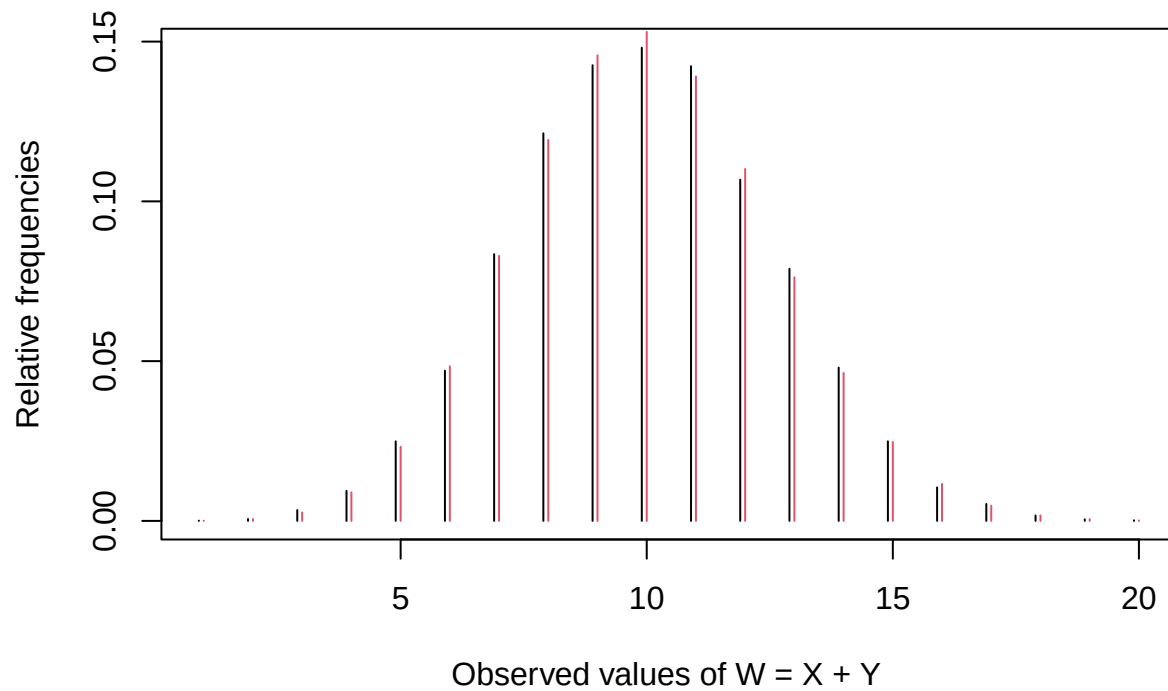
Part d)

Repeat for $B = 10,000$. Comment.

```
# 10,000 simulations  
B <- 10000  
X <- rbinom(B, m, p)  
Y <- rbinom(B, n, p)  
W <- X + Y  
table(W)
```

```
## W  
##   1    2    3    4    5    6    7    8    9   10   11   12   13   14   15   16  
##   1    6   34   94  249  470  835 1213 1426 1481 1423 1068  789  480  249  105  
##  17   18   19   20  
##  53   17    5    2
```

```
Tab = table(W)  
W.obs = as.numeric(names(Tab))  
P.obs = as.numeric(Tab/B)  
  
plot(W.obs - 0.1, P.obs, type='h',  
     xlab = "Observed values of W = X + Y",  
     ylab = "Relative frequencies")  
points(0:(m+n), dbinom(0:(m+n), m + n, p=1/3), type='h', col=2)
```



By increasing the number of trials in the simulation, the experimental values have become increasingly close to the theoretical values as seen by the minimal difference between the red (theoretical) and black (experimental) lines in the graph.